Decentralized Remote Care Transformation: A Low-Carbon, High-Equity Delivery Model for Nova Scotia

A STRATEGIC BUSINESS CASE

To: Director of Planning and Investment, Department of Health and Wellness
From: Brianna Holmes
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Subject: Capital Allocation Recommendation for Provincial Remote Care Infrastructure Expansion

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1. Executive Summary & Strategic Rationale

1.1 Objective

This business case seeks authorization for a strategic capital investment of $2.02 million over a three-year lifecycle to deploy a standardized, low-carbon, decentralized remote patient monitoring (RPM) and virtual care infrastructure network across Nova Scotia. The initiative targets complex chronic disease cohorts—specifically Congestive Heart Failure (CHF), Chronic Obstructive Pulmonary Disease (COPD) and Diabetes—within the Northern, Western, and Eastern health zones.

By scaling up secure, interoperable remote monitoring technologies, this framework aims to reduce avoidable emergency department (ED) visits by 22%, decrease acute care lengths of stay (LOS) for chronic respiratory admissions by 15%, and eliminate an estimated 140 metric tons of patient-travel carbon emissions annually by Year 3, compounding to 1,400 metric tons annually by Year 10.

1.2 Problem Statement

The Nova Scotia health system is experiencing acute capacity constraints driven by an aging demographic, a high provincial burden of chronic disease, and deep-seated geographic health disparities between the Central Zone (Halifax Regional Municipality) and peripheral rural communities. Frontline nursing structures are facing unprecedented clinical workloads, exacerbated by inefficient, reactive care delivery models that rely heavily on centralized, acute care facilities.

Simultaneously, the status quo framework for care delivery generates significant environmental externalities through patient travel and fossil-fuel-reliant clinic operations,

misaligning with the province's legislative mandates for green infrastructure and sustainable public spending. Without a structural pivot toward proactive, decentralized, data-driven remote care, the system will face escalating operational deficits and worsening wait times.

1.3 Alignment with the Nova Scotia Action Plan for Health

This proposal directly operationalizes core pillars of the Department of Health and Wellness's mandate:

1. **Becoming a Magnet for Healthier Communities:** By shifting care delivery from acute hospital beds directly into the homes of rural Nova Scotians, this initiative improves historical health equity imbalances, ensuring that residents of the Northern or Eastern zones achieve equivalent clinical access to those in the Central zone.
2. **Infrastructure Optimization and Sustainability:** The program utilizes clean-tech solutions and remote monitoring pipelines to bypass the need for expensive physical brick-and-mortar clinic expansions, utilizing data visualization architectures to optimize clinical resources.
3. **Frontline Empowerment:** By introducing automated asynchronous triage data, the model relieves burdens on acute care emergency departments, allowing nurses to operate at the top of their clinical scope via targeted, planned interventions rather than continuous reactive crisis management.

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2. Clinical Workflow & Resource Impact Analysis

2.1 Nurse-Driven Care Model and Scope Maximization

To maximize system throughput and optimize operational expenditures, this initiative implements an integrated, nurse-driven chronic disease triage hub. Rather than relying on high-cost, low-availability physician networks for routine oversight, the model utilizes the full professional competencies of Licensed Practical Nurses (LPNs), Registered Nurses (RNs), and Nurse Practitioners (NPs) working in synchronization:

- **Licensed Practical Nurses (LPNs):** Act as the operational backbone for patient onboarding and logistics. LPNs monitor the asynchronous data ingestion streams for the target cohorts (Diabetic blood-glucose trends, COPD oxygen saturation/spirometry, and CHF daily weights). They manage peripheral medical device distribution to rural participants and flag initial biometric anomalies.

- **Registered Nurses (RNs):** Serve as the primary clinical gatekeepers. RNs manage the automated alert system, evaluating complex, compounding clinical indicators (e.g., a diabetic patient experiencing concurrent respiratory distress). RNs coordinate care plan escalations and conduct targeted clinical coaching sessions via secure video channels.
- **Nurse Practitioners (NPs):** Function as the autonomous clinical leads for the regional hubs. Operating under their full legislative scope of practice in Nova Scotia, NPs interpret advanced longitudinal trends, adjust medication regimens, order necessary diagnostic laboratory workups, and sign off on acute intervention protocols, actively preventing unnecessary travel to regional emergency rooms.

2.2 Target Populations & Rural Quality Outcomes

By limiting the program scope strictly to the "big three" rural chronic disease clusters, the initiative targets the exact demographics responsible for systemic bed-blocking:

- **Congestive Heart Failure (CHF):** Daily weight tracking captures early fluid retention markers, allowing for rapid diuretic adjustments before patients progress to acute pulmonary edema.
- **Chronic Obstructive Pulmonary Disease (COPD):** Asynchronous pulse oximetry tracking identifies hypoxic trends early, enabling home-based oxygen titration or early antibiotic intervention.
- **Diabetes (Type 2 / Insulin-Dependent):** Continuous and flash glucose monitoring tracking flags dangerous HbA1c trajectories, protecting rural patients from severe glycemic events and long-term microvascular complications.

2.3 Epidemiology and Target Cohort Densities

To justify a regional hub deployment across the Northern, Western, and Eastern health zones, the program targets the three highest drivers of chronic disease morbidity in rural Nova Scotia. Based on extrapolated provincial disease registries and Canadian Chronic Disease Surveillance System data, the active target patient population across these three rural zones is modeled as follows:

- **Isolated Diabetes Mellitus Cohort:** An estimated 24,500 rural patients live with diagnosed Type 1 or Type 2 diabetes requiring clinical oversight.
- **Isolated COPD Cohort:** Approximately 14,200 rural patients diagnosed with chronic obstructive pulmonary disease.

- **Isolated Congestive Heart Failure (CHF) Cohort:** Approximately 8,800 patients under active management for structural cardiac deficits.
- **The Comorbidity Multiplier (High-Risk Sub-Cohort):** A critical 4,600-patient segment exhibits overlapping diagnoses (e.g., concurrent COPD and CHF, or brittle Diabetes paired with advanced cardiovascular disease). This comorbidity multiplier represents less than 10% of total volume but historically drives over 45% of unplanned regional hospital admissions.

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3. Financial Modeling & Total Cost of Ownership

3.1 Enterprise Software Cost-Avoidance Framework

A critical financial advantage of this model is its compliance with provincial cost-containment directives. This framework assumes a zero-dollar marginal software footprint by anchoring operations entirely within Nova Scotia's existing enterprise software ecosystems:

1. **Clinical Charting & Data Ingestion:** All patient monitoring records, telemetry flags, and longitudinal care summaries are routed directly through the province's newly deployed One Person One Record (OPOR) architecture.
2. **Virtual Engagement & Meetings:** Secure clinical consultations, interdisciplinary triage huddles, and patient-facing educational seminars utilize the provincially licensed Microsoft 365 Suite (specifically Microsoft Teams for Healthcare).

Because licensing fees, cybersecurity audits, and server maintenance for these platforms are already absorbed within central provincial information technology budgets, this project avoids millions in auxiliary vendor licensing overhead.

3.2 Human Resource Cost Projection (NSNU Matrix)

Personnel expenditures are modeled directly on hourly wage baselines derived from the Nova Scotia Nurses' Union (NSNU) collective agreement. Calculations assume standard full-time equivalencies (FTE) of 2,080 hours per annum, loaded with a 20% benefit and payroll tax factor.

The baseline deployment hub squad comprises 3 LPNs, 2 RNs, and 1 NP:

- **LPN Personnel Allocation:** 3 FTE × $36.00/hr × 2,080 hrs × 1.20 = $269,568.00 annually.
- **RN Personnel Allocation:** 2 FTE × $48.00/hr × 2,080 hrs × 1.20 = $239,616.00 annually.
- **NP Personnel Allocation:** 1 FTE × $65.00/hr × 2,080 hrs × 1.20 = $162,240.00 annually.

Total Combined Annual Hub Labor: **$671,424.00**
Total Projected 3-Year Pilot Labor Expenditure: **$2,014,272.00**

3.3 Capital Infrastructure & Hardware Requirements

Because software costs are completely avoided, the healthcare-side hardware profile is remarkably lean and sustainable. Each member of the clinical hub team requires a standard, enterprise-grade clinical laptop equipped with a secured provincial virtual private network (VPN) portal to access OPOR and Microsoft Teams remotely.

- **Hardware Allocation:** 6 Corporate-Grade Laptops × $1,500.00 per unit = **$9,000.00** (Fixed Year 1 Capital Cost).

3.4 Summary Financial Ledger (Phase 1)

Cost Center	Year 1	Year 2	Year 3	Total Lifecycle Cost
Clinical Personnel (NSNU Matrix)	$671,424.00	$671,424.00	$671,424.00	**$2,014,272.00**
Hardware (Secure Laptops)	$9,000.00	$0.00	$0.00	**$9,000.00**
Enterprise Software (OPOR/Teams)	*Amortized ($0)*	*Amortized ($0)*	*Amortized ($0)*	**$0.00**

3.5 Emergency Department Presentation Cost-Modeling & Net-Savings Runway

An average non-admitted chronic disease Emergency Department (ED) presentation in regional Nova Scotia facilities (e.g., Cape Breton Regional or Colchester East Hants Health Centre) generates a baseline public-payer expense of **$380.00 per visit** (including initial nursing triage, lab diagnostics, and basic imaging). If the presentation escalates to an acute inpatient admission with a typical 5.2-day length of stay (LOS), the baseline cost spikes to an average of **$6,800.00 per admission**.

Adjusted Status Quo Breakdown for 450 Patients

The $3,006,000 annual baseline cost for this specific 450-patient roster is driven by their unmanaged acute health patterns:

- <u>The Comorbidity Multiplier Sub-Cohort (45% of Costs / $1,352,700):</u> Consists of 45 hyper-high-risk patients (10% of the roster) with overlapping conditions (e.g., CHF

+ COPD). They average roughly 4 inpatient admissions each per year
$45 \times 4 \times \$6{,}800 = \$1.22M$

- The Isolated Chronic Clusters (55% of Costs / $1,653,300): Consists of the remaining 405 patients with single-disease profiles. Their costs are driven by frequent emergency room presentations and occasional brief hospitalizations.

Gross Annual System Costs vs. Remote Care Savings Runway with 450 Patient Caseload:

Financial Metric	**Year 1 (Pilot Launch)**	**Year 2 (Stabilization)**	**Year 3 (Full Optimization)**
Status Quo Cohort Cost	$3,006,000	$3,006,000	$3,006,000
Hub Operational Cost	$680,424 *(incl. laptops)*	$671,424	$671,424
Managed Footprint Cost	$2,555,100 *(15% reduction of Status Quo)*	$2,254,500 *(25% reduction of Status Quo)*	$1,953,900 *(35% reduction of Status Quo)*
Net System Savings	**$229,524**	**$79,076**	**$380,676**

3.6 Longitudinal Strategic Scaling Projections (10-Year Horizon)

To align with the Treasury Board's long-term capital planning cycles, the infrastructure, laptop lifecycle, and nursing headcount scale over a decade to meet projected demographic changes:

- **Year 1 (Pilot Launch):** A **Total Operational Cost of $680,424** covers initial hub operations. Proactive monitoring avoids approximately **235 acute inpatient bed days** (equivalent to preventing ~45 hospitalizations based on a 5.2-day average length of stay).
- **Year 2 (Pilot Stabilization):** A **Total Operational Cost of $671,424** maintains core operations. Refined nursing triage optimization scales bed-blocking prevention to **~390 avoided acute bed days** annually.
- **Year 5 (Regional Expansion):** A **Total Operational Cost of $2,014,272** expands the network to 3 decentralized regional hubs. This structural shift frees up **~1,560 physical bed days** per year across rural community hospitals.
- **Year 8 (System Stabilization):** A **Total Operational Cost of $2,177,000** includes a standard IT hardware refresh cycle. Automated anomaly filtering protects regional wards by intercepting chronic flare-ups, resulting in **~2,600 avoided acute bed days**.
- **Year 10 (Full Provincial Maturity):** A **Total Operational Cost of $3,500,000** achieves full provincial integration. By managing 15,000 active participants at home, the program eliminates **~13,000 acute bed days** annually.

Financial Component	**Year 1 (Pilot)**	**Year 2 (Pilot)**	**Year 5 (Expansion)**	**Year 8 (Stabilization)**	**Year 10 (Maturity)**
Active Patient Roster	450	450	1,800	3,000	15,000
Status Quo Cohort Cost	$3,006,000	$3,006,000	$12,024,000	$20,040,000	$100,200,000
Total Operational Cost	$680,424	$671,424	$2,014,272	$2,177,000	$3,500,000
Managed Footprint Cost	$2,555,100	$2,254,500	$9,018,000	$13,026,000	$65,130,000
Net Annual Savings	**$229,524**	**$79,076**	**$991,728**	**$4,837,000**	**$31,570,000**
Avoided Acute Bed Days	**~235 Days**	**~390 Days**	**~1,560 Days**	**~2,600 Days**	**~13,000 Days**

Strategic Fiscal Leverage

While the program's operational investment remains tight and flat due to the laptop-only setup and fixed nursing squad footprints, the return on investment expands significantly over the 10-year horizon. This single, localized nurse-driven chronic triage hub absorbs a highly disproportionate share of systemic risk. Replicating this low-carbon, asset-light digital framework across adjacent clinical specialties provides the Department of Health and Wellness with a clear, verified pathway to contain escalating structural expenditures while directly enhancing frontline service quality and patient care continuity.

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4. Performance Measurement & Evaluation Framework

4.1 Departmental Pillar Weighting and Program Alignment

To evaluate program performance objectively, key success indicators are weighted directly against the Nova Scotia Department of Health and Wellness's capital allocation priorities:

[Clinical Outcomes: 35%] + [Fiscal Value: 35%] + [Equity & Parity: 15%] + [Environmental Sustainability: 15%] = Total Score (100%)

4.2 Weighted SMART Goals & Quality of Life (QoL) Standards

1. **Clinical Quality Goal (Weight: 35%):** Achieve a sustained **25% reduction in avoidable all-cause 30-day readmission rates** for the registered Diabetic, COPD, and CHF patient

populations in the Eastern, Northern, and Western zones within 18 months of program enrollment.

2. **Fiscal Accountability Goal (Weight: 35%):** Secure a **net positive return on investment within 14 months** of launch, demonstrating an incremental cost-effectiveness ratio (ICER) below $30,000 per Quality-Adjusted Life Year (QALY) saved.

3. **Geographic Equity Goal (Weight: 15%):** Ensure that **at least 65% of all active enrollees reside outside the Central Zone**, targeting communities located more than 50 kilometers away from a regional secondary care hospital.

4. **Patient Care Quality of Life Standard (Weight: 15%):** Maintain a clinically meaningful mean score improvement across all active participants within 18 months of program enrollment, utilizing validated, disease-specific metrics tailored to each core tracking cohort:

    a) **Congestive Heart Failure (CHF) Cohort:** Achieve a mean reduction of **≥12 points** on the *Minnesota Living with Heart Failure Questionnaire (MLHFQ)*, indicating reduced clinical anxiety and enhanced physical tolerance.

    b) **COPD Cohort:** Achieve a mean reduction of **≥4 points** (the established minimal clinically important difference) on the *St. George's Respiratory Questionnaire (SGRQ)*, demonstrating improved self-management efficacy and breathing baseline control.

    c) **Diabetes Cohort:** Achieve a mean reduction of **≥8 points** on the *Problem Areas in Diabetes (PAID)* scale, directly measuring a reduction in diabetes-related emotional distress and an increase in daily self-care compliance.

    These synchronized metrics directly measure the overall standard of patient care, verifying that high-risk rural Nova Scotians experience an enhanced, stable quality of life while remaining safely within their home communities.

4.3 Sustainability Risk and Mitigation Matrix

To ensure successful delivery across diverse health zones, potential project vulnerabilities have been proactively modeled. The table below outlines the primary operational, environmental, and technological risks alongside targeted strategic mitigation protocols.

RISK CATEGORY	RISK DESCRIPTION	RISK LEVEL	TARGETED MITIGATION PROTOCOL
ENVIRONMENTAL & INFRASTRUCTURE	Rural broadband or cellular connectivity deficits in peripheral Northern or Eastern zones preventing real-time data transmission.	**Moderate**	Implement store-and-forward caching protocols on peripheral diagnostic equipment. Prioritize asynchronous cellular-backed telemetry devices over pure broadband-dependent configurations.
DATA PRIVACY & GOVERNANCE	Security breaches or unauthorized access to sensitive Protected Health Information (PHI) during remote ingestion or virtual consultations.	**High**	Enforce end-to-end encryption for all remote pipelines. Route all video consultations strictly through provincially managed, HIPAA/PHIA-compliant Microsoft Teams for Healthcare channels. Audit access logs automatically within the OPOR core architecture.
FRONTLINE ADOPTION	Clinical resistance or alert fatigue among hub LPNs and RNs due to high volumes of unrefined incoming biometric data streams.	**Moderate**	Integrate automated anomaly thresholds within the script to filter out baseline data noise. Establish clear, standardized operational triage paths to prevent workflow bottlenecks.
SUPPLY CHAIN CONTINUITY	Hardware depreciation, battery failures, or component shortages for rural peripheral medical devices (e.g., glucometers, spirometers).	**Low**	Establish a centralized equipment recycling and support loop managed by hub LPNs. Maintain a rolling 15% hardware buffer stock at regional hubs to ensure instant replacement availability for participants.

4.3 Sustainability Risk and Mitigation Matrix

Implementing an advanced technological deployment across rural networks introduces distinct operational and systemic risks. To safeguard public assets, uphold absolute standards of patient care, and remain compliant with the Personal Health Information Act (PHIA) of Nova Scotia, the following targeted mitigation frameworks are operationalized:

1. Rural Infrastructure & Connectivity Deficits

- **Risk Profile:** Patients residing in structurally isolated pockets of the Eastern or Western zones may suffer from intermittent cellular coverage or a complete deficit of residential broadband networks, disrupting the real-time transmission of critical biometric data streams.
- **Mitigation Protocol:** Peripheral tracking hardware deployed to rural participants utilizes dual-sim, carrier-agnostic LTE modules capable of dynamically jumping across competing networks (e.g., Bell, Rogers, Eastlink). Furthermore, data processing algorithms are engineered for fault-tolerant, asynchronous buffering, securely caching telemetry on the edge device until an active, encrypted handshake is re-established with the hub.

2. Protected Health Information (PHI) Security and Privacy Breaches

- **Risk Profile:** As synchronous and asynchronous data pipelines expand beyond centralized physical firewalls into residential environments, the threat surface for malicious intercept, device theft, or accidental disclosure of highly sensitive patient charts increases exponentially.
- **Mitigation Protocol:** All data transitions utilize end-to-end AES-256 encryption wrapped in a mandatory, hardware-enforced provincial Virtual Private Network (VPN) layer. Devices feature remote-wipe capability triggered automatically upon physical tampering or credential failure detection. No localized storage of raw clinical files occurs on patient-side modules; instead, data maps securely directly into the centralized, audited One Person One Record (OPOR) ecosystem.

3. Frontline Nursing Alert Fatigue and System Burnout

- **Risk Profile:** The continuous ingestion of clinical tracking data from thousands of active Diabetic, COPD, and CHF participants can generate massive volumes of minor biometric telemetry alerts, inadvertently overloading the core nursing huddle and causing severe data clutter or missed clinical signals.
- **Mitigation Protocol:** The dashboard pipeline utilizes customized multi-tiered baseline thresholds. Minor, isolated fluctuations in a single data stream (such as a temporary,

asymptomatic glucose surge) trigger passive, automated text alerts or reminders directly to the patient via self-management templates. The core RN and LPN triage streams are preserved exclusively for compounding alerts (e.g., sudden weight gain paired with dropping oxygen levels), allowing frontline teams to maximize their immediate clinical impact.

4. Device Supply Chain and Distribution Shortages

- **Risk Profile:** Global hardware constraints or component manufacturing delays could stall the distribution of critical testing modules (such as wireless spirometers or continuous glucose monitors), bottlenecking new patient intake and delaying regional scale milestones.
- **Mitigation Protocol:** The Department of Health and Wellness institutes a decentralized vendor pooling framework, multi-sourcing all peripheral equipment across diverse, certified alternative suppliers rather than relying on a single distributor contract. This strategy maintains an operational buffer stock of approximately 15% surplus inventory housed securely within localized regional hubs to ensure uninterrupted patient deployment.

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5. Conclusion & Next Steps

This decentralized remote care model offers the Nova Scotia Department of Health and Wellness a clear, asset-light mechanism to address escalating rural chronic disease expenditures. By combining existing enterprise software licensing (OPOR and Microsoft 365) with a specialized, nurse-driven care framework, the project delivers an estimated **$31.50 million in annual net savings by Year 10** while enhancing patient care quality and advancing the province's low-carbon infrastructure mandates.

Upon capital authorization, the immediate 90-day implementation timeline will focus on finalizing the CSV ingestion schemas, onboarding the core hub nursing staff, and initiating patient enrollment within the priority comorbidity sub-cohorts.